NUCLEO-H563ZI Pin Plan and Power Notes

STM32H563ZIT6 motor-controller board | Prepared 2026-10-20 | Sources: ST UM3115 (MB1404) user manual, STM32H563ZI datasheet

1. Board basics

Item	Value
Board / MCU	NUCLEO-H563ZI (board reference MB1404), STM32H563ZIT6
Core, memory	Cortex-M33, 2 MB flash, 640 KB SRAM, 144-pin package
Logic voltage	3.3 V by default (1.8 V by jumper JP4). All pins used here are 5 V-tolerant inputs per the datasheet; outputs are 3.3 V
USB	CN1: ST-LINK port (USB-C) for programming and serial console. CN13: user port (USB-C, sink only) for data
LEDs	LD1 green (PB0), LD2 yellow (PF4), LD3 red (PG4)
Buttons	B1 blue user button (PC13), B2 black reset
Headers	Zio headers (CN7 to CN10) are female on top. Morpho headers (CN11, CN12) are male-pin footprints, unsoldered by default
Barrel jack	None

2. Power options

Source	Voltage	Current limit	Jumper JP2
ST-LINK USB (CN1)	5 V	500 mA	pins 1-2 (default)
VIN (CN8 pin 15)	7 to 12 V	800 mA at 7 V; 450 mA at 7-9 V; 250 mA at 9-12 V	pins 3-4
E5V (CN11 pin 6)	4.75 to 5.25 V	1.3 A	pins 5-6
Charger input	5 V	not specified	pins 7-8
User USB-C (CN13)	5 V	3 A	pins 9-10
3V3_EXT (CN8 pin 7)	3.0 to 3.6 V	1.3 A (ST-LINK is unpowered)	no jumper

Recommended: a certified wall adapter into a barrel-jack module that steps down to 5 V, wired to E5V (CN11 pin 6) and ground, with JP2 on pins 5-6. Size the module for at least 2 A. Motors get their own supply; never run motor current through the Nucleo.

3. Pin plan (confirmed in manual and datasheet)

Function	Peripheral	Pins (TX / RX / extra GPIO)	Connector position
Motor 1	USART2	PD5 / PD6 / PG0	CN9 pins 6, 4, 29
Motor 2	LPUART1	PB6 / PB7 / PG1	CN10 pins 14, 16; CN9 pin 30
Motor 3	UART7	PE8 / PE7 / PG2	CN10 pins 18, 20; CN8 pin 14
Motor 4	USART6	PC6 / PC7 / PG3	CN7 pins 1, 11; CN8 pin 16
Link to other board	UART4	PD1 / PD0	CN9 pins 27, 25
Sensor 1	I2C1	PB8 (SCL) / PB9 (SDA)	CN7 pins 2, 4
Sensor 2	I2C2	PF1 (SCL) / PF0 (SDA)	CN9 pins 19, 21

The extra GPIO per motor (PG0 to PG3) is assumed to be a driver enable line. UART4 on PD0/PD1 uses the board's CAN pins, so CAN is not available.

4. Pins to avoid

Pins	Reason
PD8, PD9	USART3, the ST-LINK serial console
PA13, PA14	SWD debug
PA11, PA12, PB13, PB14, PA9, PA4, PG7	User USB-C port
PA1, PA2, PA7, PB15, PC1, PC4, PC5, PG11, PG13	Ethernet, if used
PC13, PB0	User button and green LED

5. Wiring and operating notes

- **Board-to-board link:** a 4-wire cable: 5 V, ground, TX, RX (TX and RX crossed). Power both boards from the same 5 V supply with a common ground, rather than through the Nucleo's 5 V pin, which has no published current rating.
- **Power order:** power the board from the wall first, wait for the green LD5 light, then plug the computer into the ST-LINK port (CN1). Skipping this order can damage the PC, limit its current, or leave the board unpowered.
- **Supply quality:** the manual recommends supplies that meet SELV and EN-60950-1, so use a certified wall adapter.
- **Sensors:** I2C buses may need 4.7 kOhm pull-ups to 3.3 V if the sensor modules do not include them. Power only low-current sensors from the board's 3.3 V pin.
- **5 V logic on the other board:** the Nucleo's pins accept 5 V inputs, but its outputs are 3.3 V only. A level shifter may be needed to be read reliably.

6. Open items

- Motor driver and sensor models are not yet chosen.
- The manual does not say whether the user USB-C port's 5 V line is isolated from the board supply. Check the MB1404 schematic before connecting a computer to CN13 while on wall power.
- Verify the final pin choices once in STM32CubeMX before wiring.